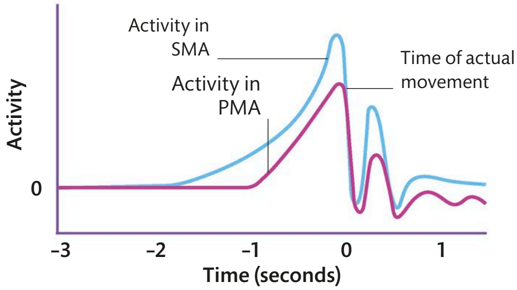
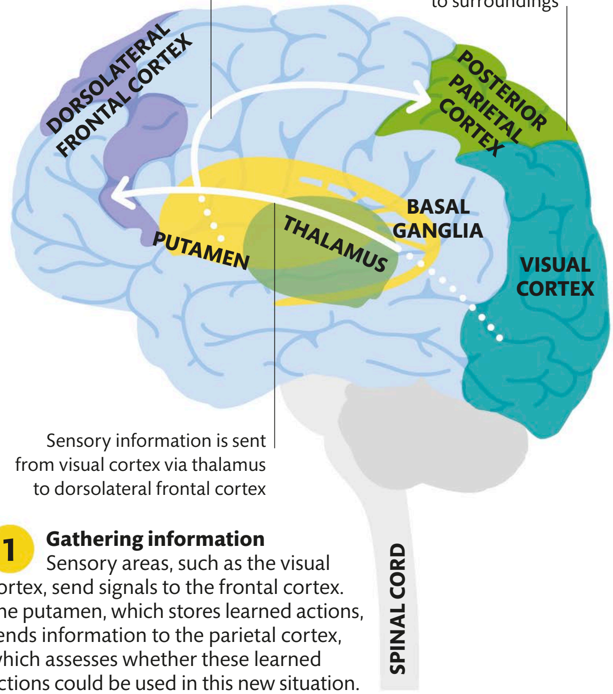
READINESS POTENTIAL

When we prepare for a voluntary action, a buildup of electrical activity, called the readiness potential, occurs. It begins in the SMA and is intensified by activity in the PMA. Activity in the SMA starts up to 2 seconds before we become consciously aware of our decision to move—which may suggest that we are less in control of our actions than we might believe (see p.168).



THE **CEREBELLUM** CONTAINS MORE THAN **50 PERCENT** OF THE BRAIN'S **NEURONS**



1 Gathering information
Sensory areas, such as the visual cortex, send signals to the frontal cortex. The putamen, which stores learned actions, sends information to the parietal cortex, which assesses whether these learned actions could be used in this new situation.

Planning Movement

Conscious movements are those that we deliberately decide to make. They involve several regions of our brain and include processes that lie outside our conscious awareness.

The planning process

There are several stages involved in carrying out a movement—from initial perception of the environment, to planning, to adjustments during the movement. These stages involve different areas of the brain working together to produce a response. The area that prompts the movement is the motor cortex. Different sections of the motor cortex send signals to different parts of the body (see p.98). However, before an action begins, an action plan is created by the dorsolateral frontal cortex and the posterior parietal cortex and is passed through two areas of the motor cortex: the supplementary motor area (SMA) and the premotor area (PMA). The cerebellum coordinates the movement as it is happening. The steps above show the brain areas involved and the sequence of signals in a typical movement.

WHY DON'T WE FORGET HOW TO RIDE A BICYCLE?

Nerve cells in the putamen encode the sequence of muscle movements into our long-term memory storage so that they are easily accessible even years later.